EXP.NO:	10	PAGE REPLACEMENT: FIFO AND LRU
DATE:		

AIM

To implement FIFO (First In First Out) and LRU (Least Recently Used) page replacement algorithms and compute total page faults and page hits.

PROCEDURE

1. Input the reference string and number of frames.
2. FIFO: Replace the page that was loaded into memory the earliest.
3. LRU: Replace the page that has not been used for the longest period.
4. Count and display total page faults and page hits.

PROGRAM 1 – FIFO Page Replacement

```
#include <stdio.h>
int main() {
    int n, frames, i, j;
    int pageFaults=0, pageHits=0;
    printf("Enter number of pages in reference string: "); scanf("%d",&n);
    int ref[n];
    printf("Enter the reference string: ");
    for(i=0;i<n;i++) scanf("%d",&ref[i]);
    printf("Enter number of frames: "); scanf("%d",&frames);
    int frame[frames];
    for(i=0;i<frames;i++) frame[i]=-1;
    int index=0;
    for(i=0;i<n;i++) {
        int found=0;
        for(j=0;j<frames;j++) if(frame[j]==ref[i]) { found=1; pageHits++; break; }
        if(!found) { frame[index]=ref[i]; index=(index+1)%frames; pageFaults++; }
    }
    printf("\nTotal Page Faults: %d\n", pageFaults);
    printf("Total Page Hits:   %d\n", pageHits);
    return 0;
}
```

PROGRAM 2 – LRU Page Replacement

```
#include <stdio.h>
int main() {
    int n, frames, i, j;
    int faults=0, hits=0, counter=0;
    printf("Enter number of pages in reference string: "); scanf("%d",&n);
    int ref[n];
    printf("Enter the reference string: ");
    for(i=0;i<n;i++) scanf("%d",&ref[i]);
    printf("Enter number of frames: "); scanf("%d",&frames);
    int frame[frames], time[frames];
    for(i=0;i<frames;i++) { frame[i]=-1; time[i]=0; }
    for(i=0;i<n;i++) {
        int found=0;
        for(j=0;j<frames;j++) {
            if(frame[j]==ref[i]) { hits++; counter++; time[j]=counter; found=1; break; }
        }
        if(!found) {
            int lru=0;
            for(j=1;j<frames;j++) if(time[j]<time[lru]) lru=j;
            counter++; frame[lru]=ref[i]; time[lru]=counter; faults++;
        }
    }
    printf("\nTotal Page Faults: %d\n", faults);
    printf("Total Page Hits:   %d\n", hits);
    return 0;
}
```

OUTPUT

```
--- FIFO ---
$ gcc fifo.c && ./a.out
Enter number of pages in reference string: 7
Enter the reference string: 1 3 0 3 5 6 3
Enter number of frames: 3

Total Page Faults: 6
Total Page Hits:   1

--- LRU ---
$ gcc lru.c && ./a.out
Enter number of pages in reference string: 7
Enter the reference string: 1 3 0 3 5 6 3
Enter number of frames: 3

Total Page Faults: 5
Total Page Hits:   2

--- Comparison (same reference string: 1 3 0 3 5 6 3, frames=3) ---
Algorithm   Page Faults   Page Hits
FIFO              6            1
LRU               5            2
```

RESULT

FIFO and LRU page replacement algorithms were successfully implemented. LRU produced fewer page faults than FIFO for the same reference string, demonstrating its superior performance by keeping recently used pages in memory.